Optimizing PCB Layout

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In a previously published article, [1], the impact of parasitics on performance was discussed. The eGaN® FET outperformed lower voltage rated MOSFETs by combining low figure of merit, low package parasitics, and low loop inductance. For the eGaN FET, the PCB layout dominated the parasitics when employing a conventional PCB Layout. This white paper will explore the optimization of PCB layout for an eGaN FET based point of load (POL) buck converter, comparing the conventional designs and proposing a new optimal layout to further reduce parasitics. The optimal layout will provide improved efficiency, faster switching speeds, and reduced device voltage overshoot compared to conventional designs. The eGaN FET based POL buck converters operate at a switching frequency of 1 MHz, an input voltage range of 12-28 V, an output voltage of 1.2 V, and an output current up to 20 A.

INTRODUCTION

eGaN FETs are available in a Wafer Level Chip-Scale Package (WLCSP) with terminals in a Land Grid Array (LGA) format. Some of these devices do not offer a separate gate-return source pin, but rather a number of very low inductance LGA solder bars as shown in figure 1. These parts can be treated in the same way as one provided with a dedicated gate return pin or bar, by allocating the source pads closest to the gate to act as the "star" connection point for both the gate loop and power loop. The layout of the gate and power loops are then separated by having the currents flow in opposite or orthogonal directions as shown in figure 1

By interleaving the drain and source terminals on one side of the device, a number of small loops with opposing currents are generated that will decrease the overall inductance through magnetic field self-cancellation. This is not only true for the PCB traces shown in figure 2(a), but also for the vertical LGA solder bars and the interlayer connection vias shown in figure 2(b). With multiple small magnetic field cancelling loops formed, the total magnetic energy, and therefore inductance, is significantly reduced. A further reduction in partial loop inductance is possible by bringing both drain and source currents out on both sides of the device from the centerline and duplicating the magnetic field cancellation effect. This works by reducing the current in each conductor, thus further reducing the energy stored, and the shorter current path yields a lower inductance.

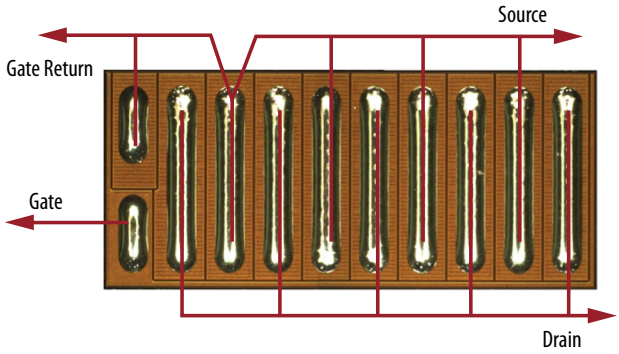


Figure 1: GaN transistor in an LGA format showing the direction of device current flow that minimizes common source inductance.

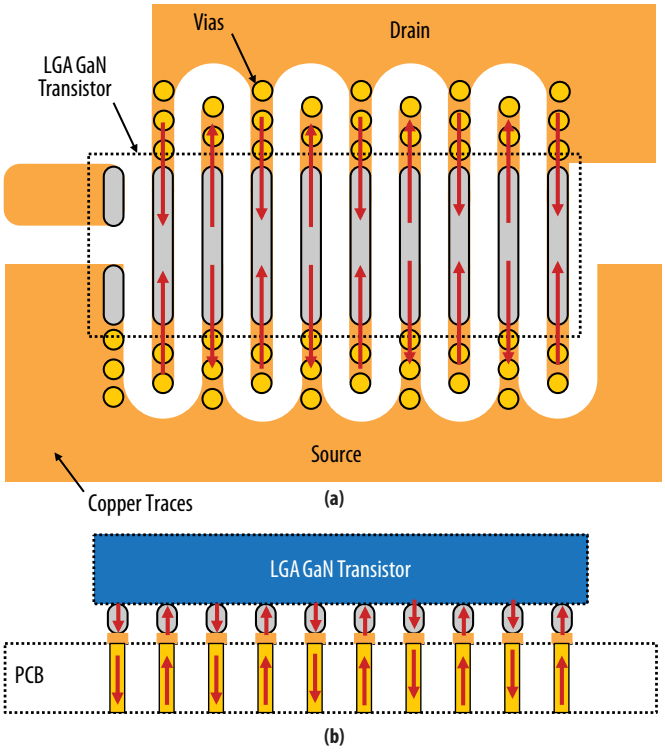


Figure 2: LGA GaN transistor mounted on a PCB showing alternating current flow in red (a) top view (b) side view.

With the significant reduction in package parasitics provided by the eGaN FET, the common source inductance is minimized and is no longer the major parasitic loss contributor. The high frequency loop inductance, controlled by PCB layout becomes the major contributor to loss, making layout of the eGaN FETs critical to high frequency performance. To verify this, different layouts with similar common source inductance and different high frequency loop inductances are compared and methods to reduce loop inductance by PCB layout are proposed.

From the efficiency curves obtained from experimental prototypes, shown in figure 3a, the impact of layout on efficiency can be seen for the eGaN FET at 1 MHz. An increase in the high frequency loop inductance from around 0.4 nH to 2.9 nH results in additional loss, decreasing efficiency by over 4%. Another disadvantage of high frequency loop inductance is that voltage overshoot increases with loop inductance. Decreasing the high frequency loop inductance results in lower voltage overshoot, increased input voltage capability, and reduced EMI. Figures 3b and 3c show the switching node waveforms for designs with a high frequency loop inductance of 1.6 nH and 0.4 nH; the voltage overshoot is reduced from 100% of the input voltage to 30%, respectively.

COMPARISON OF CONVENTIONAL PCB LAYOUTS FOR eGaN FETS LATERAL POWER LOOP

The first conventional PCB layout places the input capacitors and devices on the same side of the PCB in close proximity to minimize the size of the high frequency loop [2]. The high frequency loop for this design is contained on the same side of the PCB and is considered a lateral power loop as a result of the power loop flowing on the board plane on a single layer. An eGaN FET design arranged in a lateral power loop was created and the part placement and high frequency power loop are shown in figure 4 with the high frequency loop highlighted in red. For this design, the inductor connection is made through internal layers using vias in between the top switch and synchronous rectifier. The driver is located in close proximity to the eGaN FETs to minimize common source inductance and keep the common source inductance constant between designs. This allows a comparison of only the influence of high frequency loop inductance.

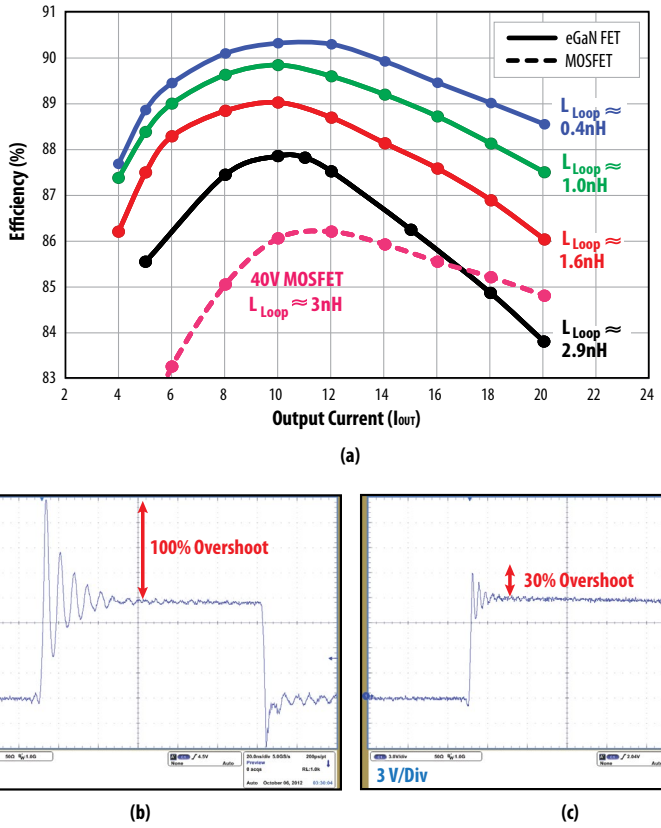


Figure 3:
(a) Impact of high frequency loop inductance on efficiency for designs with similar common source inductance
(b) Switching node waveforms of eGaN FET designs with L_{LOOP}≈1.6 nH and
(c) L_{LOOP}≈0.4 nH (V_{IN}=12 V, V_{OUT}=1.2 V, F_{SW}=1 MHz, L_{BUCK}=150 nH, eGaN FETS: Top Switch: EPC2015 Synchronous Rectifier: EPC2015, MOSFETs: Top Switch: BSZ097N04LSG Synchronous Rectifier: BSZ040N04LSG)

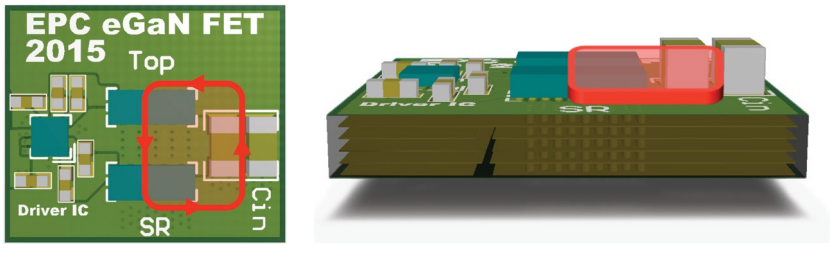


Figure 4: Conventional lateral power loop with eGaN FETs (a) Top view (b) Side view

While minimizing the physical size of the loop is important to reduce parasitic inductance, the design of the inner layers is also critical. For the lateral power loop design the first inner layer serves as a "shield layer" [3]. This layer has a critical role to shield the circuit from the fields generated by the high frequency power loop. The power loop generates a magnetic field that induces a current, opposite in direction to the power loop, inside the shield layer. The current in the shield layer generates a magnetic field to counteract the original power loop's magnetic field. The end result is a cancellation of magnetic fields that translates into a reduction in parasitic inductance at the cost of increased eddy current losses within the shield. Having a complete shield plane in close proximity to the power loop provides the best performance.

For the lateral power loop design, the high frequency loop inductance should show little dependence on board thickness as the power loop is contained on the top layer. The lateral design should be very dependent on the distance from the power loop to the shield layer which is contained on the first inner layer [4]. To minimize loop inductance in the lateral power loop, the distance from the power loop and shield layer must be minimized.

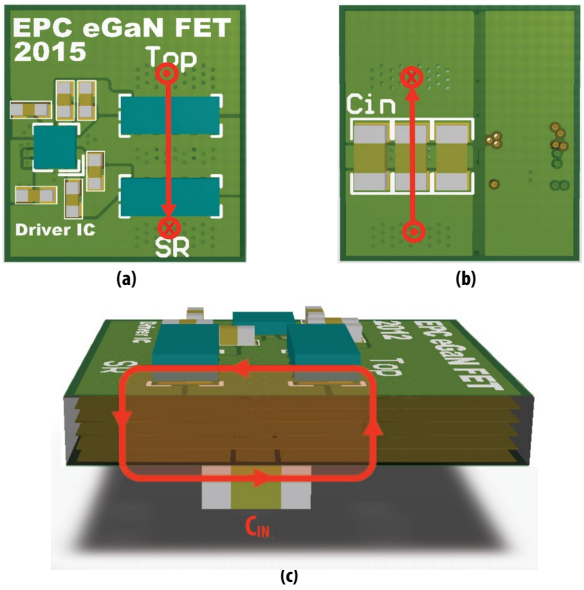


Figure 5: Conventional vertical power loop with eGaN FETs (a) Top view (b) Bottom view (c) Side view

	Lateral Loop	Vertical Loop	Optimal Loop
Single Sided PCB Capability	Yes	No	Yes
Field Self Cancellation	No	Yes	Yes
Inductance Independent of Board Thickness	Yes	No	Yes
Shield Layer Required	Yes	No	No

Table I: Characteristics of conventional and optimal PCB layout power loop designs

VERTICAL POWER LOOP

The second conventional PCB layout places the input capacitors and devices on opposite sides of the PCB, with the capacitors generally being located directly underneath the devices to minimize the physical loop size (figure 5). This layout is considered a vertical power loop because the power loop travels perpendicular to the board plane with vias connecting the power loop through the board. An eGaN FET design arranged in a vertical power loop was created and the part placement and high frequency power loop are shown in figure 5 with the high frequency loop highlighted in red. Again, space is left between the devices to allow the inductor connection.

For the vertical power loop design, there is no shield layer due to the vertical structure of the power loop. As opposed to the use of a shield plane, the vertical power loop uses a self-cancellation method to reduce inductance. For the PCB layout, the board thickness is generally much thinner than the horizontal length of the traces on the top and bottom side of the board. As the thickness of the board decreases, the area of the loop shrinks significantly when compared to the lateral power loop, and the current flowing in opposing directions on the top and bottom layers begins to provide field self-cancellation, further reducing parasitic inductance.

In the vertical power loop design, the loop inductance is heavily dependent on the board thickness as the power loop is contained on the top and bottom layers of the PCB. Without the requirement of a shield layer, the distance between the first inner layer and the power loop has little impact on the inductance. To minimize loop inductance in the vertical power loop, the board thickness must be minimized.

OPTIMAL eGaN FET LAYOUT FOR REDUCED PARASITICS

To enable the high switching speed available from the superior FOM, eGaN FETs were developed in advanced land grid array (LGA) packages that not only have low internal inductance, but enable user to design ultra-low inductance into their board. To provide the benefits of reduced loop size, magnetic field self-cancellation, consistent inductance independent of board thickness, a single sided PCB design, and high efficiency for a multi-layer structure, an improved layout is proposed for eGaN FETs. The design utilizes the first inner layer, shown in figure 6b, as a power loop return path. This return path is located directly underneath the top layer's power loop, figure 6a, allowing for the smallest physical loop size combined with field self-cancellation. The side view (figure 6c) illustrates the concept of creating a low profile self-cancelling loop in a multilayer PCB. The characteristics of the conventional and proposed optimal designs are compared in table I.

The improved layout places the input capacitors in close proximity to the top device, with the positive input voltage terminals located next to the drain connections of the top eGaN FET. The eGaN FETs are located in the same positions as the lateral and vertical power loop cases. Located between the two eGaN FETs is a series of interleaved switching node and ground vias arranged to match the LGA fingers of the synchronous rectifier eGaN FET. The

interleaved switching node and ground vias are duplicated on the bottom side of the synchronous rectifier. These interleaved vias provide three advantages:

(1) The via set located in between the two eGaN FETs provides a reduced length high frequency loop inductance path leading to lower parasitic inductance.

(2) The via set located beneath the synchronous rectifier eGaN FET provides additional vias for reduced resistance during the synchronous rectifier eGaN FET freewheeling period, reducing conduction losses.

(3) The interleaving of the via sets with current flowing in opposing direction allows for reduced eddy and proximity effects, reducing AC conduction losses.

EXPERIMENTAL RESULTS

To compare the performance of the proposed optimal power loop with conventional lateral and vertical designs for a wide range of applications, four separate board builds were created. The designs varied the overall thickness of the board and the distance between the top layer and the first inner layer in the board (inner layer distance). The part layouts remained unchanged (shown in figures 4, 5, and 6), and all designs were comprised of four layers with two ounce copper thickness.

The values of the high frequency loop inductance for varying board thicknesses and inner layer distance were simulated and the results are presented in figure 7b. From the data it can be seen that for the lateral power loop the board thickness has little impact on the high frequency loop inductance while the inner layer distance (the distance from the power loop to the shield layer) significantly impacts the inductance. For the vertical power loop, the inner layer distance has very little impact on the inductance of the design, while the board thickness significantly impacts the inductance by as much as 80% when the board thickness is doubled from 31 to 62 mils.

For the proposed optimal layout, the design shares the traits of the lateral power loop by showing little dependence on board thickness and a strong dependence on inner layer distance. This design provides a significant reduction in loop inductance from the removal of the shield layer and reduced physical size of the power loop; traits similar to the vertical power loop design. Combining the strengths of both conventional designs, and limiting the weaknesses, the proposed design can provide a reduction in inductance on the order of 65% compared to the best conventional lateral or vertical power loop.

The power loss for the three different loop layouts, constructed with different board thicknesses and inner layer distances is shown in figure 8. From this data it can be seen that for similar parasitic inductances the power loss of the lateral loop is higher than the vertical and optimal loop. The cause of the increased loss in the lateral power loop can be attributed to the additional loss in the shield layer, which is not required in the vertical or proposed optimal power loop. The experimental hardware verifies the predicted trend of increased loop inductance and higher power loss.

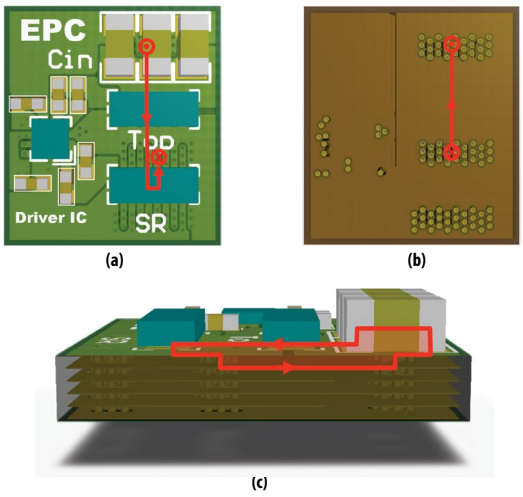


Figure 6: Proposed optimal power loop with eGaN FETs
(a) Top view
(b) Top view of inner layer 1 (c) Side view

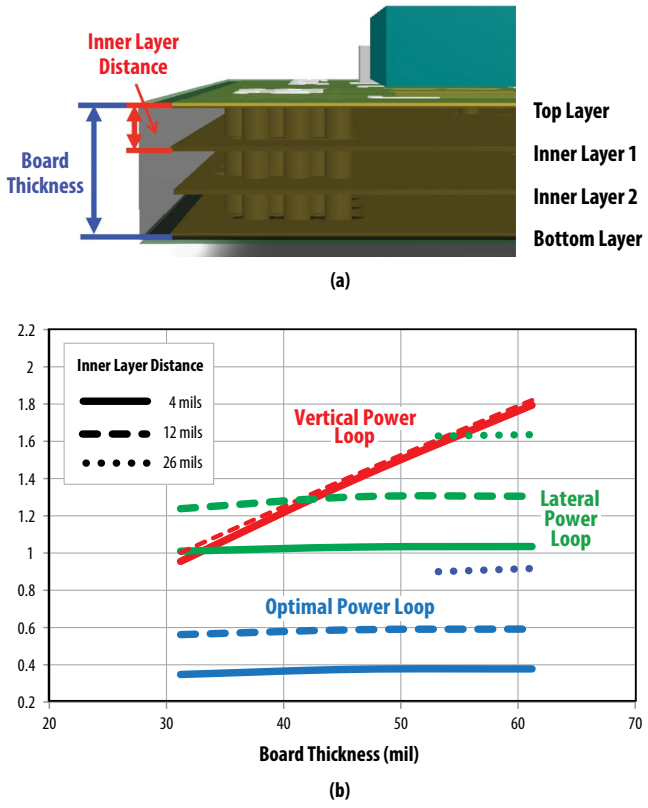


Figure 7:
(a) PCB cross section drawing of board thickness and inner layer distance for experimental designs
(b) Simulated high frequency loop inductance values for lateral, vertical, and optimal power loops with different board thickness and inner layer distance

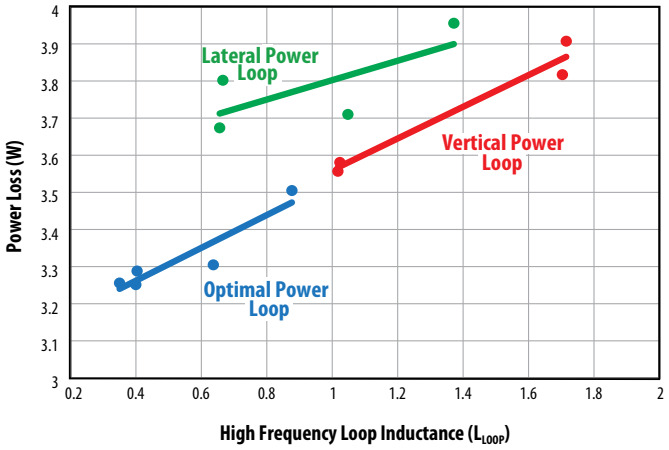


Figure 8: Experimental power loss plot for lateral, vertical, and optimal power loop designs. (V_{IN}=12 V, V_{OUT}=1.2 V, I_{OUT}=20 A, F_{SW}=1 MHz, L_{BUCK} =300 nH, Top Switch: EPC2015 Synchronous Rectifier: EPC2015)

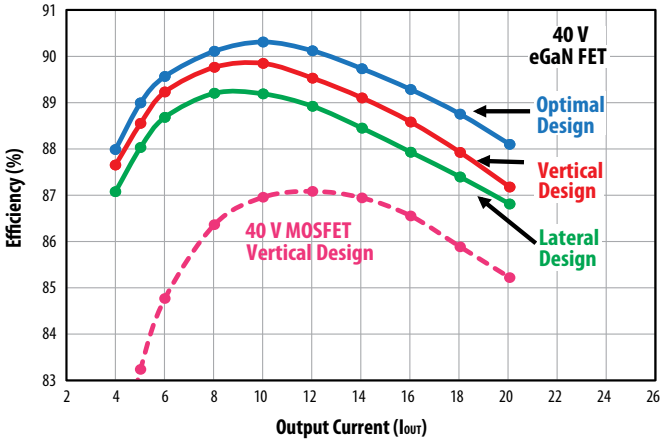


Figure 9: Efficiency comparisons for different loop designs (V_{IN}=12 V, V_{OUT}=1.2 V, F_{SW}=1 MHz, L_{BUCK} =300 nH, Board Thickness=31 mils, Inner Layer Distance=4 mils, eGaN FETs: Top Switch: EPC2015, Synchronous Rectifier: EPC2015, MOSFETs: Top Switch: BSZ097N04LSG, Synchronous Rectifier: BSZ040N04LSG)

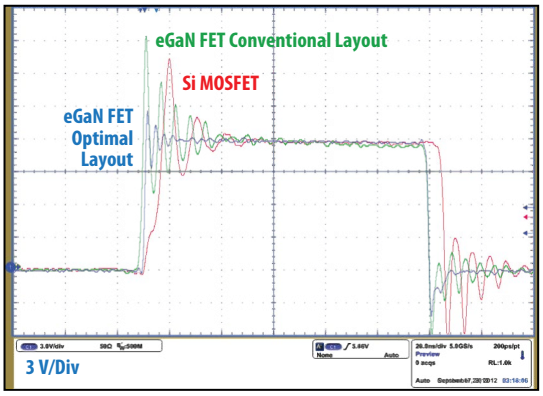


Figure 10: Switching node waveforms of optimal eGaN FET, conventional eGaN FET, and MOSFET designs at V_{IN}=12 V (V_{OUT}=1.2 V, I_{OUT}=20 A, F_{SW}=1 MHz, L_{BUCK} =300 nH, 40 V eGaN FETs: Top Switch: EPC2015, Synchronous Rectifier: EPC2015, MOSFETs: Top Switch: BSZ097N04LSG, Synchronous Rectifier: BSZ040N04LSG)

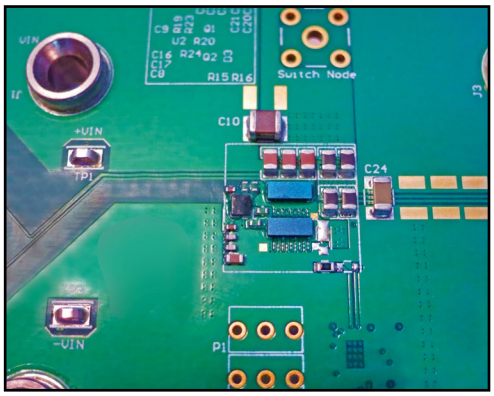


Figure 11: EPC9107 development board using optimal design (Top Switch: EPC2015, Synchronous Rectifier: EPC2015, Driver: LM5113)

Shown in figure 9 are the efficiency results of the three proposed designs compared to a silicon implementation utilizing a vertical power loop with the smallest commercial package, a 3x3mm TSDSON-8, to minimize the power loop. For the Si MOSFET design, the high frequency loop inductance was measured to be around 2 nH, compared to 1 nH for a similar power loop using eGaN FETs. This is due to the large packaging inductance of the Si MOSFET dominating the loop design. As a result of the superior FOM and packaging of the eGaN FETs, all of the power loop structures outperform the Si MOSFET benchmark design. With the optimal power loop, the efficiency is significantly improved for the eGaN FETs. A 3% full load efficiency improvement is achieved when compared to the Si MOSFET.

For the different eGaN FET designs, the optimal power loop provides a 0.8% and 1% full load efficiency improvement over the vertical and lateral power loops, respectively. For all of the design tests, the optimal layout provides the highest efficiency and lowest device voltage overshoot.

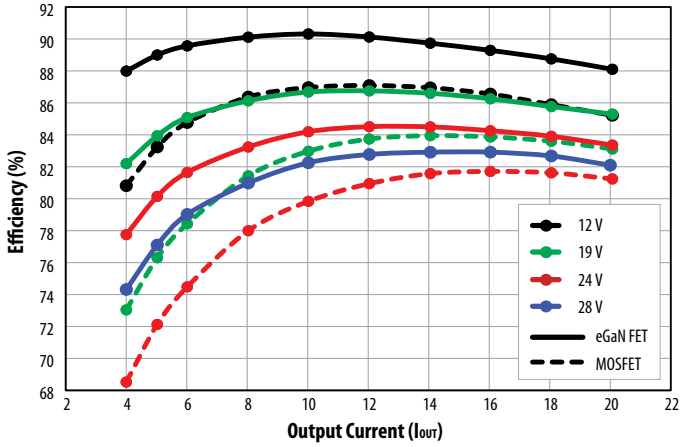
The switching waveforms for the eGaN FET conventional and optimal layouts and Si MOSFET benchmark are shown in figure 10. Both eGaN FET designs offer significant switching speed gains when compared to the Si MOSFET benchmark. For the eGaN FET with the conventional vertical layout, the high switching speed combined with loop inductance induces a large voltage spike. The optimal layout eGaN FET offers a 500% increase in switching speed with a 40% reduction in voltage overshoot when compared to the

40 V Si MOSFET benchmark. For the eGaN FET with low package parasitic inductance, the layout is critical to switching at high speeds, limiting device overshoot, and improving efficiency.

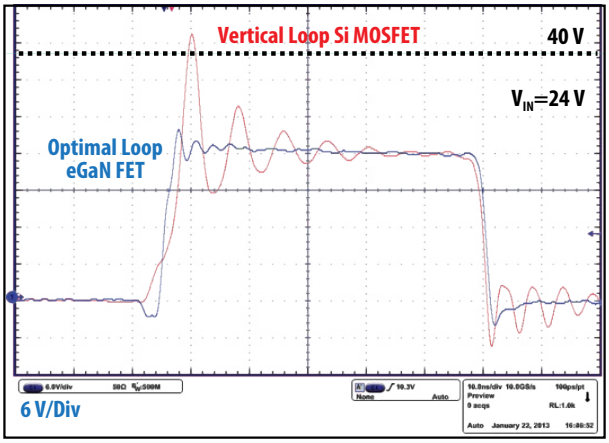
With the reduced overshoot and high efficiency achievable with the optimal eGaN FET layout, the converter has the ability to handle much higher input voltages with low voltage rated devices. The converter was operated at input voltages of 12 V, 19 V, 24 V, and 28 V, and the efficiency curves are shown in figure 10. Due to the voltage overshoot in the MOSFET design, its operation was limited to 12 V, 19 V, and 24 V.

The introduction of high performance eGaN FETs offers the potential to switch at higher frequencies and efficiency than possible with traditional Si MOSFET technology. Combined with improved figures of merit and low parasitic packaging, eGaN FETs enable extremely low high frequency loop inductance layout to fully utilize the device's capability. To evaluate the impact of high frequency loop inductance on performance, multiple designs of conventional lateral and vertical power loops with the same minimal common source inductance were created and compared.

To overcome the limitations of the conventional PCB layouts an optimal layout is proposed to achieve the best performance with eGaN FETs. Through the use of an optimal layout approach, the benefits of eGaN FET technology are further enhanced, providing additional efficiency gains and higher voltage operation capability.



(a)



(b)

Figure 12:
(a) Efficiency comparison of eGaN FET optimal layout and MOSFET at varying input voltages
(b) Switching node waveforms at V_{IN}=24 V and I_{OUT}=20 A (V_{OUT}=1.2 V, F_{SW}=1 MHz, L_{BUCK}=300 nH, eGaN FETs: Top Switch: EPC2015, Synchronous Rectifier: EPC2015, MOSFETs: Top Switch: BSZ097N04LSG, Synchronous Rectifier: BSZ040N04LSG)

References:

[1] D. Reusch, "eGaN-Silicon Power Shoot-Out Vol. 13, Part 1: Impact of Parasitics on Performance" Power Electronics Technology, March 2013.
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